Piush Kole

ML Research & Applied Machine Learning | Model Compression · Reproducible Benchmarking · Scientific Computing

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SUMMARY

MSc Data Science candidate with hands-on research experience in neural network quantization and reproducible experimental design, built through an independent benchmark evaluating post-training and quantization-aware training for a high-energy-physics jet-classification task. Comfortable owning a project end to end: experimental design, implementation, statistical evaluation, automated testing, and public documentation. Seeking ML research, applied ML, or scientific-computing internship roles.

TECHNICAL SKILLS

- **ML / quantization:** PyTorch, post-training quantization (PTQ), quantization-aware training (QAT), mixed-precision search, scikit-learn, statistical evaluation (paired CIs, multi-seed design)
- **Scientific computing / HEP tooling:** hls4ml (PyTorch → fixed-point C-simulation), NumPy, pandas
- **Software engineering:** Python, pytest (automated regression + API test suites), Git, FastAPI, Docker, CI (GitHub Actions)
- **Data & visualization:** SQL, Power BI, Tableau, Plotly, Matplotlib
- **Other:** AWS (Cloud Practitioner), LangChain/LangGraph (agentic LLM systems)

RESEARCH PROJECT — HEP-TRIGGER QUANTIZATION LAB (INDEPENDENT RESEARCH)

github.com/Piush2004/hep-trigger-quantization-lab — public repository, code + reports + reproduction instructions

An independent, reproducible benchmark of low-bit neural-network quantization for a simulated high-energy-physics jet-classification task (not CERN-affiliated; built on a public, particle-level Monte Carlo dataset, not real detector data).

- Designed and ran a multi-seed (5-seed) benchmark comparing FP32, INT8/INT6/INT4 PTQ, and QAT across two MLP architectures, with pre-registered selection rules to prevent test-set leakage; one exploratory result that *was* test-set-selected was identified, disclosed as biased, and excluded from every confirmatory claim rather than hidden.
- Ran a near-exhaustive mixed-precision search (2,187 layouts; every combination of INT4/INT6/INT8 across 7 quantizable weight/activation units) and found a QAT-selected layout that reliably beats uniform INT6 PTQ at 78% of its storage (all 5 seeds, 95% CI [0.0011, 0.0015] ROC-AUC).
- Benchmarked the quantized networks against non-neural baselines (logistic regression, gradient-boosted trees) and reported, rather than omitted, that a default-hyperparameter GBM beats the FP32 MLP on raw accuracy — framing the neural network's actual case as FPGA deployability, not accuracy.
- Implemented a PyTorch → fixed-point (`hls4ml`) C-simulation conversion path and validated it against the full-precision model: strong agreement at sufficient bit-width (≥97% class agreement, W≥14), and a real accuracy collapse found at narrow width under default arithmetic — substantially mitigated by rounding and saturation, reported as found rather than smoothed over. Trained a machine-learning surrogate for FPGA resource estimation on the public `wa-hls4ml` dataset — explicitly labelling every number by evidence tier (measured / simulated / estimated / not available) so simulated and measured results are never conflated.
- Built a FastAPI + Plotly research dashboard exposing the frozen results, covered by 14 automated API and security tests (biased-result exclusion, CORS/secret checks, device-separation invariants); packaged with Docker and a PWA manifest for public deployment.
- Engineering discipline: 141 automated tests passing across the data pipeline, quantization code, hardware-estimation path, and API layer (two independent runs, byte-identical results); every reproduction command and dataset fingerprint documented in `docs/REPRODUCIBILITY.md`.
- Wrote up the full study as a research-style manuscript (methodology, statistics, limitations, reproducibility statement) — see `paper/` in the repository.

SECOND RESEARCH PROJECT — SYNTHETIC SILENCE

Independent research prototype, in progress — mapping information voids & epistemic risk in generative-AI responses

- Built a pipeline estimating epistemic risk in LLM responses: extracting atomic claims, retrieving supporting literature live (OpenAlex), and combining evidence-density, sparse-autoencoder reconstruction error, and linguistic specificity/uncertainty signals into a continuous, explainable risk score — explicitly documented as a heuristic, not a calibrated truth classifier.
- Iterated through three architecture versions after live multi-LLM validation surfaced real bugs (false positives on true claims, a false negative on a fabricated one); each failure documented and fixed rather than hidden, with before/after evidence in the repository's own validation report.
- 100/100 backend tests and 15/15 extension tests passing; CI covers the test suite, an extension syntax check, and a Docker build check.

EDUCATION

M.Sc. Data Science (Computer Science) — CHRIST (Deemed to be University), Bengaluru · *Jul 2025 – May 2027 (in progress)*

B.Sc. (Hons.) Data Science — Techno India University, Kolkata · *May 2022 – Apr 2025* · 69.67%

EXPERIENCE

Data Executive Intern — Cybertoss Pvt. Ltd., Kolkata (onsite) · *Jun 2024 – May 2025*

- Owned the end-to-end data lifecycle (collection, cleaning, validation, reporting) for large-scale datasets feeding downstream analytics and model-ready pipelines.
- Automated data-processing workflows in Python, reducing manual effort and improving delivery speed for cross-functional analytical projects.

Data Science Trainee — RemarkSkill, via KSHITIJ (IIT Kharagpur's techno-management festival), remote · *Jun 2024*

- Built hands-on proficiency in Python (pandas, NumPy, Matplotlib) and Power BI through applied dataset projects and data-storytelling deliverables.

CERTIFICATIONS

Agentic AI in Action: RAG with LangChain (Skillsoft, 2025) · AWS Cloud Practitioner Essentials (AWS Academy, 2025) · Introduction to Business Analysis Using Spreadsheets (Coursera, 2025) · Python for Data Science & Machine Learning: Zero to Hero (Udemy, 2023)

LEADERSHIP

IEEE Committee Member, CHRIST University · *Dec 2025* — coordinated outreach to 10+ industry companies for the IEEE student conference as part of a 20-member organizing committee.

Other experience (full-stack/agentic LLM application development, business-analytics projects) is available on request; this CV is scoped to research and ML-engineering evidence most relevant to scientific-computing and ML-research roles.